Version 1.1

Physics PhD Atlas 2027

The International Applicant's Playbook (US, Canada + Europe)

Version 1.1 · program data checked October 22, 2026 · 82 programs (36 US, 16 Canada, 30 Europe: Denmark, Finland, France, Germany, Ireland, Italy, Netherlands, Spain, Sweden, Switzerland, United Kingdom)

This playbook is the "how" that goes with the Atlas data. Every program-specific fact below comes from the Atlas, where each value links to an exact quote on the program's official page. General advice is labelled as advice. Nothing here is immigration, visa or legal advice. For those topics, use only official government sources (see Section 13).

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1. Start here (10 minutes)

1. Open **Atlas.html** in any browser. It works offline. Filter by what blocks you most (Physics GRE, English-test waiver, fee waivers), then star programs to build a shortlist.
2. Open **Atlas.xlsx** (Excel, Google Sheets, or LibreOffice). Pick your starred programs in the **Shortlist** tab. Deadlines, fees and rules fill in automatically. The **Budget**, **Recommenders** and **Supervisor outreach** tabs do the rest of the tracking.
3. Read Sections 2 to 7 of this playbook today. They cover the decisions with deadlines attached: fee waivers, English tests, GRE scores, Canadian supervisors, and Europe's different system.
4. Before you pay any fee or book any test, click the value in the Atlas and open the official page. Programs change rules mid-cycle. When the Atlas says *Not stated*, the pages we checked did not say. It does **not** mean "no".

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2. Your remaining timeline (late-October start)

Treat these as working targets (advice). The fixed dates come from the programs.

When	Do this
This week	Shortlist 8 to 15 programs. Check which need the **Physics GRE** (Section 5) and whether your English-test plan covers every program (Section 4).

When	Do this
By **Nov 1, 2026**	If Rice is on your list, request its worldwide **Doctoral Application Fee Waiver Weeks** waiver; Rice says it then needs a completed application by **Nov 27**.
Oct 31 – Nov 3	Europe's first wave. EPFL's autumn round closes **Oct 31** (next round Apr 30). Heidelberg's HGSFP portal and IMPRS Quantum Dynamics close **Nov 1**. Imperial's first President's PhD Scholarship round closes **Nov 2**. IMPRS Astronomy (Heidelberg) closes **Nov 3**.
Early November	Email Canadian supervisors (Section 6), and the European professors or advertised positions you want (Section 7). Send recommenders their packet (Section 10).
Nov 01: Heidelberg (HGSFP); Nov 01: IMPRS Quantum Dynamics (Heidelberg); Nov 02: Imperial College London; Nov 03: IMPRS Astronomy (Heidelberg)	The earliest deadlines in this Atlas. Several programs review applications the moment the deadline passes.
Nov 30	IMPRS Quantum Materials (Dresden) closes for its February selection workshop.
Mid-December	Most US deadlines fall on Dec 11-16 (see the deadline table). Cambridge's funding deadline is **Dec 8**; Durham's is **Dec 15** for full scholarship consideration.
January	Most Canadian deadlines (Jan 7-15), plus UIUC, Penn State, Texas A&M and Rutgers. Bristol's funding deadline: applications must arrive before **Jan 18**. Edinburgh's physics funding deadline is **Jan 22**.
Feb-April	Interviews and offers. Several programs in this Atlas (e.g., Stony Brook and Texas A&M) ask admitted students to respond by **April 15**. SISSA's spring call closes in early March; EPFL's second round on Apr 30.

Deadlines in this Atlas (earliest first)

Deadline	Program	Country	Note
Nov 01, 2026	Heidelberg (HGSFP)	Germany	year inferred; act by 2026-11-01
Nov 01, 2026	IMPRS Quantum Dynamics (Heidelberg)	Germany	act by 2026-11-01
Nov 02, 2026	Imperial College London	United Kingdom	act by 2026-11-02
Nov 03, 2026	IMPRS Astronomy (Heidelberg)	Germany	act by 2026-11-03
Nov 30, 2026	IMPRS Quantum Materials (Dresden)	Germany	act by 2026-11-30
Nov 30, 2026	Ohio State	USA	year inferred
Dec 01, 2026	Alberta	Canada	year inferred
Dec 01, 2026	University of Arizona	USA	
Dec 01, 2026	UBC	Canada	
Dec 01, 2026	University of Colorado Boulder	USA	year inferred

Deadline	Program	Country	Note
Dec 01, 2026	UT Austin	USA	year inferred
Dec 01, 2026	Vanderbilt University	USA	
Dec 03, 2026	Columbia	USA	
Dec 08, 2026	Cambridge	United Kingdom	
Dec 11, 2026	Maryland	USA	
Dec 11, 2026	Toronto	Canada	year inferred
Dec 14, 2026	Stanford	USA	
Dec 14, 2026	UChicago	USA	
Dec 15, 2026	Boston University	USA	year inferred
Dec 15, 2026	Caltech	USA	year inferred
Dec 15, 2026	Cornell	USA	year inferred
Dec 15, 2026	Durham	United Kingdom	
Dec 15, 2026	Georgia Tech	USA	year inferred
Dec 15, 2026	Harvard University	USA	
Dec 15, 2026	Indiana University Bloomington	USA	year inferred
Dec 15, 2026	Johns Hopkins University	USA	year inferred
Dec 15, 2026	Massachusetts Institute of Technology	USA	year inferred
Dec 15, 2026	McGill	Canada	year inferred
Dec 15, 2026	Purdue	USA	year inferred
Dec 15, 2026	Rice	USA	
Dec 15, 2026	Stony Brook	USA	year inferred
Dec 15, 2026	Berkeley	USA	
Dec 15, 2026	University of California, Irvine	USA	year inferred
Dec 15, 2026	University of California, Los Angeles	USA	
Dec 15, 2026	Minnesota	USA	year inferred
Dec 15, 2026	University of Notre Dame	USA	year inferred
Dec 15, 2026	University of Pittsburgh	USA	
Dec 15, 2026	University of Virginia	USA	year inferred
Dec 15, 2026	University of Washington (Seattle)	USA	
Dec 15, 2026	UW-Madison	USA	
Dec 15, 2026	Yale University	USA	

Deadline	Program	Country	Note
Dec 16, 2026	UCSB	USA	
Jan 01, 2027	Rutgers	USA	year inferred
Jan 04, 2027	North Carolina State University	USA	
Jan 07, 2027	Queen's	Canada	year inferred; rolling
Jan 08, 2027	University of Manitoba	Canada	year inferred
Jan 09, 2027	Waterloo	Canada	year inferred
Jan 10, 2027	Texas A&M	USA	
Jan 15, 2027	Carleton University	Canada	year inferred
Jan 15, 2027	Penn State	USA	year inferred
Jan 15, 2027	SFU	Canada	year inferred; rolling
Jan 15, 2027	Calgary	Canada	year inferred
Jan 15, 2027	UIUC	USA	
Jan 17, 2027	Bristol	United Kingdom	
Jan 22, 2027	Edinburgh	United Kingdom	
Feb 01, 2027	McMaster	Canada	year inferred; rolling
Feb 01, 2027	University of Guelph	Canada	year inferred; sources conflict
Mar 01, 2027	Western University	Canada	
Apr 01, 2027	Dalhousie University	Canada	year inferred
Apr 30, 2027	EPFL	Switzerland	year inferred
Not stated	Munich (MCQST)	Germany	
Not stated	SISSA	Italy	
Not stated	Ottawa	Canada	
Not stated	Padova	Italy	
Not stated	Paris-Saclay	France	
Per position	Chalmers	Sweden	
Per position	TU Delft	Netherlands	
Per position	ETH Zurich	Switzerland	
Per position	ICFO (Barcelona)	Spain	
Per position	KTH Stockholm	Sweden	
Per position	Leiden	Netherlands	
Per position	Stockholm	Sweden	

Deadline	Program	Country	Note
Per position	TU Munich	Germany	
Per position	Amsterdam (UvA)	Netherlands	
Per position	Copenhagen (NBI)	Denmark	
Per position	Zurich (UZH)	Switzerland	
Per position	Uppsala	Sweden	
Rolling	Aalto	Finland	
Rolling	IMPRS Solar System (Göttingen)	Germany	2026/27 round dates not yet published
Rolling	Trinity College Dublin	Ireland	
Rolling	Bonn-Cologne (BCGS)	Germany	
Rolling	University of Saskatchewan	Canada	

"Year inferred" means the page gave a day and month but no year; we assumed the Fall 2027 cycle. Confirm on the page.

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3. Build a balanced list, and a budget

Advice. Aim for 8 to 15 applications. Choose by research fit first and prestige second. A program where two or more faculty work on what you want is a stronger application than a famous name where nobody does.

- **Fit test:** for each program, name two faculty members and one recent paper from each that you could discuss. If you can't, it isn't a fit yet.
- **Spread:** mix programs where your profile is typical of admits with a few reaches. Use each department's own descriptions (several state typical GPAs or class sizes on their pages) rather than forum rumours.
- **Cost reality:** the median international application fee in this Atlas is **90 USD (about ₹8,500)** for US programs and **140 CAD (about ₹9,400)** for Canadian programs, where stated. Ten US applications therefore cost roughly **₹85,000** before test-score fees. Many European programs charge nothing (Section 7). Rupee figures use the Bank of Canada rate of October 21, 2026 (1 USD ≈ ₹94.5), and your bank adds a margin.
- **Budget it** in the **Budget** tab, which converts every currency to rupees with an editable rate. Add score-sending and transcript costs from the official sites; we don't estimate those for you.

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4. English tests in 2026: what changed and who waives

Verified change: TOEFL iBT moved to a new **1-6 score scale on January 21, 2026**. Programs now list two minimums, one per scale (e.g., UBC: 90 on the old scale or 4.5 on the new one). Check which scale your score is on.

There are four kinds of waiver rule. Which kind a program uses decides whether an English-medium Indian degree is enough:

1. **Friendly to an English-medium Indian degree ("Yes" in the Atlas):** Aalto, TU Delft, EPFL, Harvard University, IMPRS Astronomy (Heidelberg), IMPRS Solar System (Göttingen), McGill, Rutgers, Stanford, Stockholm, Stony Brook, TU Munich, University of Arizona, Bristol, UBC, University of Colorado Boulder, Copenhagen (NBI), University of Notre Dame, UT Austin, Toronto. Most of these look at your university's language of instruction. Some ask for an official letter or transcript notation (e.g., Toronto), and some for documents sent to the department (McGill). UT Austin is different: its exempt-country list includes India itself.
2. **Conditional / list-based.** Exemption only if your institution meets a specific test. Alberta publishes a list that includes the IITs (all campuses), IISc and the University of Hyderabad but **not** the IISERs. Waterloo accepts only institutions listed on WHED with English as the *only* language of instruction, and explicitly rejects medium-of-instruction letters. Queen's lets you *request* an exemption after a recent year of English-medium study. In this Atlas: Chalmers, Columbia, Dalhousie University, KTH Stockholm, IMPRS Quantum Materials (Dresden), McMaster, Queen's, Trinity College Dublin, Alberta, University of California, Irvine, Edinburgh, University of Guelph, University of Saskatchewan, University of Virginia, University of Washington (Seattle), Waterloo, Vanderbilt University, Yale University.
3. **No waiver for an English-medium Indian degree ("No"):** Carleton University, Durham, Georgia Tech, Imperial College London, Indiana University Bloomington, Massachusetts Institute of Technology, SFU, Ohio State, Calgary, Berkeley, Cambridge, UIUC, University of Manitoba, Western University. These either use an "English-primary country" list that India is not on (e.g., UIUC, Berkeley, SFU, Calgary, and Cambridge via the UK government's list), or require a test from everyone (Georgia Tech).
4. **Not stated** on the pages checked: ask the program before you assume.

Advice: if even one program on your list needs a score, book the test now. Score delivery takes time; Waterloo, for example, says it takes 4-6 weeks for official scores to reach it. Also check speaking minimums. Several programs tie teaching-assistant eligibility (and so part of your funding) to a separate, higher speaking score.

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5. GRE and Physics GRE: required again at some top programs

Most programs in this Atlas don't need the GRE, and several will **not even look** at it. A notable group requires the **Physics GRE** for this cycle:

- **Caltech**: Physics GRE REQUIRED for the Fall 2027 cohort
- **Cornell**: REQUIRED for theory applicants (theoretical astro, particle, condensed matter); theory applications without it cannot be considered
- **Harvard University**: Physics GRE subject test required
- **Stanford**: Physics GRE REQUIRED for all applicants
- **Calgary**: REQUIRED for all foreign applicants
- **Berkeley**: Physics GRE REQUIRED for Fall 2027 entry
- **UChicago**: Physics GRE REQUIRED for the 2026-2027 cycle

- **University of Guelph**: Physics GRE required for applicants whose degrees are not from a Canadian institution (minimum 800); an application below a minimum is still considered as a whole
- **University of Virginia**: Valid Physics GRE score required (self-reported at application; official scores needed on matriculation)

Programs that will not consider **any** GRE score, even if you send one: Rice, Stony Brook, University of California, Irvine, UCSB, Minnesota, University of Notre Dame, Vanderbilt University. Programs that ignore the **General** GRE but still review the Physics GRE: UT Austin, UW-Madison.

Advice:

- If your list contains any program in the first group, you need a Physics GRE score that arrives **before that program's deadline**. Check the remaining ETS test dates and score-reporting times on ets.org today; don't rely on forum dates.
- If you have a strong score and a program marks it *Optional* or *Recommended*, sending it can help. UT Austin is a special case: it asks you to explain in your Statement of Purpose why your Physics GRE matters, or it ignores the score.
- If you have no score and your list has none of the "Required" programs, don't spend scarce weeks on it now.

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6. Canada works differently: supervisors and master's-first rules

Supervisor first

In Canada, admission often depends on a professor agreeing to take you. In this Atlas:

Program	Supervisor before admission	What the page says
Carleton University	Not stated	Not stated on the page checked
Dalhousie University	Required	Admission depends on finding a supervisor through the free pre-application/interview; the formal application is submitted only with a supervisor's support and names the supervisor
McGill	Recommended	Not required, but highly recommended: professors select students for their groups
McMaster	Recommended	Funding includes a research fellowship from the supervisor's grant; contact faculty directly to ask if they are taking students
Queen's	Recommended	A complete application needs a research proposal; work with a potential supervisor early
SFU	Not stated	Not stated in the text checked; transfers need supervisory committee approval
Alberta	Not stated	Physics pages do not state a pre-application supervisor requirement; FGPS notes some thesis programs require one

Program	Supervisor before admission	What the page says
UBC	Not required	Name faculty of interest in the application; no prior commitment needed, but contacting faculty is encouraged
Calgary	Required	No supervisor needed to apply, but you are admitted only if at least one potential supervisor will fund you; contacting faculty first helps
University of Guelph	Required	You do not need a supervisor before applying (list 2-3 potential supervisors), but approved applicants are admitted only if a supervisor match is found
University of Manitoba	Required	Department cannot admit without a formal advisor commitment; contact a prospective advisor (via the associate head) before applying, and name your preferred supervisor in the application
Ottawa	Required	Supervisor's name required AT APPLICATION; you must be accepted by a thesis supervisor to register
University of Saskatchewan	Recommended	Supervisor approval recommended before applying; applications without a confirmed supervisor are circulated to faculty
Toronto	Not required	No supervisor needed to apply; MSc and 5-year direct-entry PhD students pick one after arrival; 4-year PhD entrants (with a master's) must secure one before registration, typically by May
Waterloo	Required	Admission requires a faculty member willing to supervise you; contact before or after applying and name supervisors on the application
Western University	Required	Every admission offer is made together with the student's faculty supervisor; contacting potential supervisors before applying is strongly encouraged

Master's-first rules

Many Canadian PhD programs expect a master's degree. A BSc holder usually starts in the MSc and may transfer to the PhD later. In this Atlas, "Master's usually required": Carleton University, McGill, Queen's, SFU, UBC, University of Guelph, University of Manitoba, Ottawa, University of Saskatchewan, Waterloo, Western University. Direct or either route: Dalhousie University, Alberta, Toronto.

Important for 5-year integrated BS-MS students (IISER, IISc, IIT and others): whether your degree counts as a master's equivalent is decided by each university's credential assessment. **Ask each program in writing.** Don't assume either way. Put the question in your first email to the graduate office and keep the reply.

Funding is stated more often

Canadian programs often publish guaranteed minimums (e.g., UBC at least CAD 40,000 a year for years 1-4; Alberta CAD 100,000 over four years; McMaster at least CAD 21,000 after tuition and fees). Note whether a figure is before or after tuition.

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7. Europe works differently

The 30 European programs in this Atlas (Denmark, Finland, France, Germany, Ireland, Italy, Netherlands, Spain, Sweden, Switzerland, United Kingdom) use two very different ways in.

1. **Calls with a deadline.** You apply to a graduate school or program by a published date, often followed by interviews. Examples:
    - Nov 1: IMPRS Quantum Dynamics, and Heidelberg's HGSFP portal.
    - Nov 2: Imperial's first President's PhD Scholarship round.
    - Nov 3: IMPRS Astronomy (Heidelberg).
    - Nov 30: IMPRS Quantum Materials (Dresden).
    - Dec 8: Cambridge's funding deadline.
    - Dec 15: Durham, for full scholarship consideration.
    - Before Jan 18: Bristol's funding deadline.
    - Jan 22: Edinburgh.
    - Spring: SISSA's exam-based call.
    - Oct 31 and Apr 30: EPFL's two rounds.
2. **Apply for a paid position, or agree a supervisor first.** There is no central committee: PhD places are advertised as jobs, or a professor must accept you before you apply. Sweden (Stockholm, KTH, Chalmers, Uppsala), Denmark (Copenhagen), the Netherlands (Leiden, Amsterdam, Delft), Switzerland (ETH Zurich, Zurich), TU Munich, ICFO and Trinity College Dublin work this way. In this Atlas, position- or supervisor-based: Aalto, Chalmers, TU Delft, EPFL, ETH Zurich, ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Solar System (Göttingen), Stockholm, TU Munich, Trinity College Dublin, Bonn-Cologne (BCGS), Amsterdam (UvA), Copenhagen (NBI), Zurich (UZH), Paris-Saclay, Uppsala.

Program	Country	Deadline	Supervisor first?	Funding	Entry	Fee
Copenhagen (NBI)	Denmark	Per position	Required	Guaranteed	Master's usually required	Not stated
Aalto	Finland	Rolling	Required	Competitive	Master's usually required	Not stated
Paris-Saclay	France	Not stated	Required	Guaranteed	Master's usually required	Not stated
Bonn-Cologne (BCGS)	Germany	Rolling	Required	Not stated	Not stated	Not stated
Heidelberg (HGSFP)	Germany	Nov 01, 2026	Not required	Not stated	Master's usually required	Not stated
IMPRS Astronomy (Heidelberg)	Germany	Nov 03, 2026	Not required	Guaranteed	Master's usually required	Not stated
IMPRS Quantum Dynamics (Heidelberg)	Germany	Nov 01, 2026	Not stated	Competitive	Master's usually required	Not stated
IMPRS Quantum Materials (Dresden)	Germany	Nov 30, 2026	Not required	Guaranteed	Either	0 (no fee)

Program	Country	Deadline	Supervisor first?	Funding	Entry	Fee
IMPRS Solar System (Göttingen)	Germany	Rolling	Not required	Typical	Master's usually required	Not stated
Munich (MCQST)	Germany	Not stated	Not stated	Not stated	Not stated	Not stated
TU Munich	Germany	Per position	Required	Typical	Master's usually required	Not stated
Trinity College Dublin	Ireland	Rolling	Required	Typical	Direct from bachelor's	Not stated
Padova	Italy	Not stated	Not stated	Competitive	Master's usually required	Not stated
SISSA	Italy	Not stated	Not stated	Not stated	Not stated	Not stated
Amsterdam (UvA)	Netherlands	Per position	Required	Typical	Master's usually required	Not stated
Leiden	Netherlands	Per position	Required	Guaranteed	Master's usually required	Not stated
TU Delft	Netherlands	Per position	Required	Typical	Master's usually required	Not stated
ICFO (Barcelona)	Spain	Per position	Required	Guaranteed	Master's usually required	Not stated
Chalmers	Sweden	Per position	Not stated	Guaranteed	Master's usually required	0 (no fee)
KTH Stockholm	Sweden	Per position	Required	Guaranteed	Master's usually required	Not stated
Stockholm	Sweden	Per position	Required	Guaranteed	Master's usually required	Not stated
Uppsala	Sweden	Per position	Required	Guaranteed	Master's usually required	0 (no fee)
EPFL	Switzerland	Apr 30, 2027	Required	Guaranteed	Master's usually required	0 (no fee)
ETH Zurich	Switzerland	Per position	Required	Typical	Not stated	Not stated
Zurich (UZH)	Switzerland	Per position	Required	Typical	Master's usually required	100 CHF
Bristol	United Kingdom	Jan 17, 2027	Recommended	Competitive	Master's usually required	Not stated
Cambridge	United Kingdom	Dec 08, 2026	Recommended	Competitive	Master's usually required	20 GBP
Durham	United Kingdom	Dec 15, 2026	Recommended	Competitive	Either	Not stated

Program	Country	Deadline	Supervisor first?	Funding	Entry	Fee
Edinburgh	United Kingdom	Jan 22, 2027	Not required	Competitive	Direct from bachelor's	Not stated
Imperial College London	United Kingdom	Nov 02, 2026	Not required	Competitive	Direct from bachelor's	0 (no fee)

Fees are often low or zero.

- **No application fee:** Chalmers, EPFL, Imperial College London, IMPRS Quantum Materials (Dresden), Uppsala.
- **Small fees:** Cambridge charges £20 per applicant per year for up to 7 doctoral applications. Zurich (UZH) charges CHF 100.
- **Watch for tuition instead.** From January 2027, Copenhagen charges PhD tuition of DKK 85,000 a year, which your funding must cover.

A master's is usually expected first. In this Atlas: Aalto, Chalmers, TU Delft, EPFL, Heidelberg (HGSFP), IMPRS Astronomy (Heidelberg), ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Quantum Dynamics (Heidelberg), IMPRS Solar System (Göttingen), Stockholm, TU Munich, Amsterdam (UvA), Bristol, Cambridge, Copenhagen (NBI), Padova, Zurich (UZH), Paris-Saclay, Uppsala. Bachelor's holders can enter at: Durham, Imperial College London, IMPRS Quantum Materials (Dresden), Trinity College Dublin, Edinburgh. Other routes:

- Heidelberg (HGSFP and IMPRS Astronomy) has 4+4 tracks for excellent candidates with a 4-year bachelor's.
- EPFL accepts 4- or 5-year bachelor's holders who explain their education in the Statement of Objectives.
- Edinburgh says most of its current PhD students had no master's before starting.
- Swedish rules (KTH, Stockholm) accept 240 credits including at least 60 at advanced level, so a 4-year degree with enough advanced courses can qualify.

If you hold a 5-year BS-MS, ask each program how it assesses your degree, and keep the reply.

Funding usually means a job contract.

- **Sweden:** funding for the whole doctorate is a condition of admission, and doctoral students are usually salaried. KTH quotes SEK 33,000–37,100 a month.
- **Funded-only admission elsewhere:** Copenhagen, ICFO and Paris-Saclay admit only students who are funded.
- **Positions:** Leiden's PhD projects are 4-year paid positions, and ETH doctorates are, with few exceptions, linked to a position.
- **The UK:** funding is competitive. Imperial says studentships for overseas applicants are very restricted.

In this Atlas, paid or typically funded: Chalmers, TU Delft, EPFL, ETH Zurich, IMPRS Astronomy (Heidelberg), ICFO (Barcelona), KTH Stockholm, Leiden, IMPRS Quantum Materials (Dresden), IMPRS Solar System (Göttingen), Stockholm, TU Munich, Trinity College Dublin, Amsterdam (UvA), Copenhagen (NBI), Zurich (UZH), Paris-Saclay, Uppsala; competitive: Aalto, Durham, Imperial College London, IMPRS Quantum Dynamics (Heidelberg), Bristol, Cambridge, Edinburgh, Padova.

English rules for Indian degrees vary a lot.

- **Friendly:**
  - Aalto lists India for English-medium degrees.
  - Bristol accepts an official medium-of-instruction letter.
  - Stockholm and IMPRS Solar System count an English-taught degree.
  - TU Delft exempts English-taught master's degrees in principle.
  - IMPRS Astronomy (Heidelberg) and Copenhagen assess English at interview instead of by test.
- **School-level English counts in Sweden:** at KTH and Chalmers, your CBSE or CISCE Class XII English grades can meet the requirement.
- **Strict:** Cambridge, Imperial and Durham exempt only degrees from listed countries, and India isn't one of them. Edinburgh accepts English-taught degrees only from institutions on its approved list.

Advice.

- In position-based systems, your supervisor email (Section 8) *is* the application. Check the vacancy pages weekly; KTH, for example, announces positions nine times a year.
- Allow time after an offer for contracts and permits. EPFL says about 3 months is usually needed for visas and work permits.

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8. Supervisor emails that get answered (advice)

Find people, not just programs. Use faculty pages, the group's own website (look for "accepting students"), and papers from the last two years. Shortlist 2-4 people per university.

The 6-sentence email:

> **Subject:** Prospective PhD student (Fall 2027): [specific topic] in your group
>
> 1. Who you are: degree, institution, expected graduation.
> 2. The specific work of theirs you read, and one genuine question or connection.
> 3. Your most relevant research experience and what *you* did in it.
> 4. One skill or result that would help their group (a technique, code, measurement).
> 5. The ask: "Are you planning to take new PhD students for Fall 2027, and would you be open to my application?"
> 6. Thanks, with CV and transcript attached as PDFs named Lastname_CV.pdf.

Cadence: one polite follow-up after about 10 days if you get no reply, then move on. Log every email in the **Supervisor outreach** tab; it calculates the follow-up date for you.

Don'ts: no mass emails with swapped names, no long life stories, no attachments over a few MB, and no generic praise ("your esteemed research").

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9. Statement of purpose: a physics framework (advice)

Programs set their own length rules. Alberta, for example, wants **500 words or less on a single page**. Read each program's prompt and obey it exactly.

A structure that works for research PhDs:

1. **Opening (2-3 sentences):** the physics question that drives you, stated concretely ("how disorder changes spin dynamics in..."), not "I have loved physics since childhood".
2. **Research experience (40-50% of the text):** for each project, give the question, *your* contribution, the method, the result, and what you learned, including what went wrong and how you fixed it.
3. **Skills and preparation:** techniques, instruments, code, and advanced courses that map onto their work.
4. **Why this program:** two named faculty and why, plus facilities or centres that matter for your plans.
5. **Goals:** where the PhD leads and why this training fits.

Check before you submit: every sentence should be true and specific, and you should be able to defend it in an interview. Write it yourself. Never use purchased or ghost-written statements.

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10. Recommendation letters: the request kit (advice)

- **Who:** research supervisors first, then instructors of advanced courses who know your work. Programs differ on number: Toronto wants **exactly two** (no more), while UIUC, Columbia and Maryland list three.
- **When:** ask at least 4-6 weeks before your earliest deadline, and send portal invitations early.
- **What to send (one email, one PDF):** your shortlist with deadlines and portal links, CV, draft SOP, a "brag sheet" (projects, dates, what you did, skills shown), and a reminder of specific moments they saw your work.
- **Reminders:** a friendly note 10 days and again 3 days before each deadline. The **Recommenders** tab tracks status per program.

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11. Fee waivers, ethically

What this Atlas found for international applicants:

- **Free or waived for everyone this cycle:** Penn State. **No application fee at all:** Chalmers, EPFL, Imperial College London, IMPRS Quantum Materials (Dresden), Uppsala.
- **Open to internationals (need, hardship, or a time-boxed window):** Columbia, Cornell, Rice, University of Notre Dame, University of Pittsburgh, Yale University.
- **Limited or conditional routes:** Caltech, Georgia Tech, Massachusetts Institute of Technology, Purdue, Rutgers, Stanford, Ohio State, University of Arizona, UBC, University of California, Los Angeles, Cambridge, UChicago, University of Colorado Boulder, UT Austin, University of Virginia, Vanderbilt University.

- **Domestic applicants only:** Boston University, North Carolina State University, Stony Brook, Texas A&M, Berkeley, University of California, Irvine, UCSB, University of Washington (Seattle), UW-Madison.
- **No waivers:** Carleton University, Indiana University Bloomington, Trinity College Dublin, Alberta, Calgary, Toronto.

Advice:

- Request a waiver **before** paying. Application fees are typically non-refundable (UChicago's Physical Sciences Division says so for its $90 fee), so a fee paid too early is gone.
- Ask briefly and honestly: state your circumstances and the program you're applying to, and attach evidence if asked. Never misstate citizenship, income or event attendance.
- Some departments have small waiver budgets that run out months before the deadline (Rutgers says so), so ask early.

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12. Submission checklist

- [ ] Program-specific statement prompt answered (length, format, required sections)
- [ ] CV in PDF, with research section, publications/posters, skills
- [ ] Transcripts uploaded for every post-secondary institution (some programs, e.g., Toronto, ask for official copies only after admission)
- [ ] English score sent from the testing agency, or waiver evidence ready
- [ ] GRE / Physics GRE sent if required (check institution codes on each page)
- [ ] Recommenders invited, and each has confirmed they will submit
- [ ] Fee paid or waiver approved, with the receipt saved
- [ ] Supervisor names listed where the program asks (Waterloo, Ottawa, UBC and others)
- [ ] Final check of every value in the Atlas against the live official page

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13. Money, safety and what this kit does not cover

- **Scams:** be wary of anyone who guarantees admission, sells "special" fee-waiver codes, or asks for your portal passwords. Real waiver codes come from the program itself; Penn State, for example, publishes its own code on its admissions page.
- **Paying fees:** many portals take Visa or Mastercard credit cards. Some accept debit cards selected as "credit" (UBC says this). Enable international online transactions with your bank before the deadline week.
- **Visas and study permits:** out of scope. Use only official sources: travel.state.gov for US student visas, canada.ca (IRCC) for Canadian study permits, and for Europe the host university's international office and the national immigration authority it points you to. Paid immigration advice in Canada is a regulated activity, and this kit is not that.

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14. How this data was built (and how to report an error)

- Each of the 82 programs was checked against its own department and graduate-school pages on October 22, 2026. Every value stores the exact sentence it came from, the URL and the date. A script confirms that each quote appears verbatim on the fetched page.
- Research and drafting were AI-assisted. Every published value is tied to an official quote, and uncertain items are labelled *Not stated*, *Conditional*, *year inferred* or *sources conflict* instead of guessed.
- A weekly automated check re-reads the cited pages and flags quotes that disappear. Updates are sent to buyers as new versions at no charge.
- Some well-known programs couldn't be verified on the check date because their sites blocked automated checks or were down for maintenance. They are not in this edition: Oxford, UCL, Manchester, ISTA Austria, Vienna, Göttingen's central pages, Geneva, KU Leuven, ENS Paris (ICFP) and the University of Barcelona. They'll be added once their pages can be verified.
- **Found something wrong or changed?** Reply to your purchase email with the program and the page link. Confirmed corrections ship in the next update.